



ENERGY TRANSPORT IN COUPLED BEAMS AND PLATES

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ABSTRACT

The mathematical theory of microlocal analysis of hyperbolic partial differential equations shows that the energy density associated to their high-frequency solutions satisfies transport equations, or radiative transfer equations for randomly heterogeneous materials with correlation lengths comparable to the (small) wavelength. The main limitation to date to the existing theory is in the consideration of boundary or interface conditions for the energy and power flow densities. This paper deals with the transport regime in a heterogeneous two-beams or two-plates system. An analytical model for the derivation of high-frequency power flow reflection/transmission coefficients at a beam or a plate junction is proposed. These results may be used in subsequent computations to solve numerically the transport equations for coupled structures, including interface conditions. Applications of this research concern the prediction of the transient response of slender structures impacted by acoustic or mechanical shocks.

INTRODUCTION

Elastic structures exhibit typical transport and diffusive behaviours in the higher frequency range of vibration. Statistical energy analysis (SEA) and heat conduction analogy (HCA) are popular methods among mechanical engineers to deal with such phenomena. However these approaches are approximate, if not incorrect, and based on some very restrictive hypotheses. A rigorous derivation of the transport regime for elastic media has been proposed recently [1,3]. These results may be easily extended to slender structures such as beams, plates or shells [4]. They generalize SEA or HCA and alleviate a great deal of their inconsistency. The main shortcoming of this theory to date is in the consideration of boundary conditions for energetic observables. In this paper we recall the basic features of the proposed method and its application to thick beams and plates. Then we show how to compute formally power flow reflection/transmission coefficients for such structures, as well as boundary reflection coefficients as a particular case of the former. A numerical example is finally presented.

ENERGY ESTIMATORS AND TRANSPORT REGIME

General setting

We consider a system of partial differential equations in time $t \in \mathbb{R}_+$ and space $\mathbf{x} \in \Omega \subset \mathbb{R}^d$, $d = 1, 2$ or 3 , written in the rescaled variables $t \rightarrow t/\varepsilon$ and $\mathbf{x} \rightarrow \mathbf{x}/\varepsilon$ as:

$$\varepsilon \partial_t \mathbf{u}_\varepsilon + P(\mathbf{x}, \varepsilon D) \mathbf{u}_\varepsilon = \mathbf{0} \quad (\text{Eq. 1})$$

for the state vector $\mathbf{u} \in \mathbb{C}^n$, with $\mathbf{u}_\varepsilon(\mathbf{x}, t) = \mathbf{u}(\mathbf{x}/\varepsilon, t/\varepsilon)$. Here ε is a small arbitrary parameter that goes to 0 in the high-frequency limit (for example the wavelength) and P is a pseudo-differential operator of matrix-valued symbol $P(\mathbf{x}, \mathbf{k})$ which is smooth with respect to $\mathbf{x}$ and its dual variable (wave vector) $\mathbf{k} \in \mathbb{R}^d$:

$$P(\mathbf{x}, \varepsilon D) \mathbf{u} = \int_{\mathbb{R}^d} e^{i\mathbf{k} \cdot \mathbf{x}} P(\mathbf{x}, \varepsilon \mathbf{k}) \hat{\mathbf{u}}(\mathbf{k}) \frac{d\mathbf{k}}{(2\pi)^d}.$$

$\hat{\mathbf{u}}$ is the Fourier transform of $\mathbf{u}$ into $L^2(\mathbb{R}^d)$ defined by:

$$\hat{\mathbf{u}}(\mathbf{k}) = \int_{\mathbb{R}^d} e^{-i\mathbf{k} \cdot \mathbf{x}} \mathbf{u}(\mathbf{x}) d\mathbf{x}, \quad \mathbf{u}(\mathbf{x}) = \int_{\mathbb{R}^d} e^{i\mathbf{k} \cdot \mathbf{x}} \hat{\mathbf{u}}(\mathbf{k}) \frac{d\mathbf{k}}{(2\pi)^d}.$$

« High-frequencies » are initiated in the system (Eq. 1) by so-called « ε -oscillatory » initial conditions of the form:

$$\mathbf{u}_\varepsilon(\mathbf{x}, 0) \sim \varepsilon^{-\frac{d}{2}} \mathbf{u}_0\left(\frac{\mathbf{x}}{\varepsilon}\right), \quad \|\nabla \mathbf{u}_0\|_{L^2} < \infty.$$

The associated energy density of that system may be estimated using the Wigner transform of the state vector. The latter is defined for two functions $\mathbf{u}$ and $\mathbf{v}$ in $[L^2(\mathbb{R}^d)]^n$ by:

$$\mathbf{W}_\varepsilon[\mathbf{u}, \mathbf{v}](\mathbf{x}, \mathbf{k}, t) = \int_{\mathbb{R}^d} e^{i\mathbf{k} \cdot \mathbf{y}} \mathbf{u}\left(\mathbf{x} - \varepsilon \frac{\mathbf{y}}{2}, t\right) \otimes \overline{\mathbf{v}\left(\mathbf{x} + \varepsilon \frac{\mathbf{y}}{2}, t\right)} \frac{d\mathbf{y}}{(2\pi)^d}. \quad (\text{Eq. 2})$$

Let $\mathbf{W}_\varepsilon(\mathbf{u}_\varepsilon) = \mathbf{W}_\varepsilon[\mathbf{u}_\varepsilon, \mathbf{u}_\varepsilon]$, then it is easily seen that the energy density is linked to:

$$\|\mathbf{u}_\varepsilon(\mathbf{x}, t)\|^2 = \int_{\mathbb{R}^d} \text{Tr} \mathbf{W}_\varepsilon(\mathbf{u}_\varepsilon)(\mathbf{x}, \mathbf{k}, t) d\mathbf{k},$$

so that the trace $\text{Tr} \mathbf{W}_\varepsilon(\mathbf{u}_\varepsilon)$ is indeed a phase-space energy density at position $\mathbf{x}$ and wave vector $\mathbf{k}$. The Wigner transform of the solution $\mathbf{u}_\varepsilon$ of (Eq. 1) has the following properties [1,3]:

- i. If the sequence $(\mathbf{u}_\varepsilon)$ lies in a bounded subset of $[L^2(\mathbb{R}^d)]^n$, then $\mathbf{W}_\varepsilon(\mathbf{u}_\varepsilon) \rightharpoonup \mathbf{W}(\varepsilon \rightarrow 0)$ weak-* in $\mathcal{S}'(T^*\mathbb{R}^d)$ up to an extracted sub-sequence – still indexed by ε – and the weak limit $\mathbf{W}$ is an hermitian, positive measure called the Wigner measure of $(\mathbf{u}_\varepsilon)$.
- ii. $\mathbf{W}$ satisfies $\mathbf{P}\mathbf{W} + \mathbf{W}\mathbf{P}^* = \mathbf{0}$, $\mathbf{P}^* = \overline{\mathbf{P}}^T$ for the conjugate transpose, and thus writes:

$$\mathbf{W} = \sum_{\alpha} w_{\alpha} \mathbf{b}_{\alpha} \otimes \mathbf{b}_{\alpha}, \quad w_{\alpha} = (\mathbf{W} \mathbf{c}_{\alpha}, \mathbf{c}_{\alpha}),$$

where the $\mathbf{b}_{\alpha}$ (respectively $\mathbf{c}_{\alpha}$) are the right eigenvectors (respectively the left) of the dispersion matrix $\mathbf{L} = -i\mathbf{P}$ on $T^*\mathbb{R}^d \setminus \{(\mathbf{x}, \mathbf{k}); \mathbf{k} = \mathbf{0}\}$ with normalization $(\mathbf{b}_{\alpha}, \mathbf{c}_{\beta}) = \delta_{\alpha\beta}$; they correspond to polarization modes for the system energy.

- iii. Furthermore coefficients $w_{\alpha}(\mathbf{x}, \mathbf{k}, t)$, called specific intensities in the literature, are solutions of the Liouville equations:

$$\partial_t w_{\alpha} + \{\lambda_{\alpha}, w_{\alpha}\} = 0 \quad (\text{Eq. 3})$$

where λ_{α} is the eigenvalue associated to the mode $\mathbf{b}_{\alpha}$, and $\{f, g\} = \nabla_{\mathbf{k}} f \cdot \nabla_{\mathbf{x}} g - \nabla_{\mathbf{x}} f \cdot \nabla_{\mathbf{k}} g$ stands for the usual Poisson bracket. Here it is assumed that this eigenvalue is simple. If λ_{α} is a multiple eigenvalue of order n_{α} , specific intensities w_{α} become $n_{\alpha} \times n_{\alpha}$ matrices but their evolution properties are globally unchanged.

For isotropic elastic media eigenvalues λ_{α} have the form $\lambda_{\alpha}(\mathbf{x}, \mathbf{k}) = \pm c_{\alpha}(\mathbf{x})|\mathbf{k}|$, such that specific intensities $w_{\alpha}^{\pm}$ are solutions of (Eq. 3) and satisfy $w_{\alpha}^{-}(\mathbf{x}, \mathbf{k}, t) = w_{\alpha}^{+}(\mathbf{x}, -\mathbf{k}, t)$ corresponding to forward and backward (energy) waves with the celerity $c_{\alpha}(\mathbf{x})$, propagating along directions $\hat{\mathbf{k}}$ and $-\hat{\mathbf{k}}$, respectively, where $\hat{\mathbf{k}} = \mathbf{k}/|\mathbf{k}|$ (normalized wave vector). Once (Eq. 3) has been solved for all energy modes, the energy density of the system is computed by:

$$E(\mathbf{x}, t) = \sum_{\alpha} \int_{\mathbb{R}^d} w_{\alpha}^{+}(\mathbf{x}, \mathbf{k}, t) d\mathbf{k} \quad (\text{Eq. 4})$$

and its power flow density is computed by:

$$\boldsymbol{\pi}(\mathbf{x}, t) = \sum_{\alpha} c_{\alpha}(\mathbf{x}) \int_{\mathbb{R}^d} w_{\alpha}^{+}(\mathbf{x}, \mathbf{k}, t) \hat{\mathbf{k}} d\mathbf{k}. \quad (\text{Eq. 5})$$

Application to beams

Let us consider a thick straight beam of mass per unit length ρ , cross-section area S , inertia I , Poisson ratio ν , Young's modulus E , shear reduction coefficient $\kappa < 1$, and reduced shear modulus $G' = \kappa G$ with $G = E/2(1+\nu)$. Its axial force N , shear force T , and bending moment M are given by:

$$N = ES \partial_s u, \quad T = G'S(\partial_s v - \theta), \quad M = EI \partial_s \theta \quad (\text{Eq. 6})$$

as functions of the axial displacement u , deflection v , and cross-section bending rotation θ . Here s is the position along the beam axis. Equations of motion are:

$$\rho \partial_t^2 u = \partial_s N, \quad \rho \partial_t^2 v = \partial_s T, \quad \frac{\rho I}{S} \partial_t^2 \theta = \partial_s M + T. \quad (\text{Eq. 7})$$

Introducing $\mathbf{u} = (\partial_t u, \partial_t v, \partial_t \theta, N, T, M)$, (Eq. 6) and (Eq. 7) have the form of a system (Eq. 1) with symbol $\mathbf{P}(s, k) = i\mathbf{L}(s, k)$ and $d = 1$, $n = 6$, where:

$$\mathbf{L} = -k \begin{bmatrix} \mathbf{0} & \mathbf{L}_{12} \\ \mathbf{L}_{21} & \mathbf{0} \end{bmatrix}, \quad \mathbf{L}_{12} = \text{diag}\left(\frac{1}{\rho}, \frac{1}{\rho}, \frac{S}{\rho I}\right), \quad \mathbf{L}_{21} = \text{diag}(ES, G'S, EI). \quad (\text{Eq. 8})$$

The eigenvalues of the dispersion matrix $\mathbf{L}$ are:

$$\begin{aligned} \lambda_T^\pm &= \pm c_T |\mathbf{k}| \quad \text{each with multiplicity 1,} \\ \lambda_P^\pm &= \pm c_P |\mathbf{k}| \quad \text{each with multiplicity 2,} \end{aligned} \quad (\text{Eq. 9})$$

with $c_T^2 = G'S/\rho$ and $c_P^2 = ES/\rho$. Mode T corresponds to a pure transverse shear energy, and the double mode P corresponds to longitudinal and bending energies.

Application to plates

Let us consider a thick plane plate with mid-surface Σ , mass per unit surface ρ , thickness h , Poisson ratio ν , Young's modulus E , shear modulus $G = E/2(1+\nu)$, and shear reduction coefficient $\kappa < 1$. Its in-plane force tensor $\mathbf{N}$, bending moment tensor $\mathbf{M}$, and shear force $\mathbf{T}$ are:

$$\mathbf{N} = h\mathbf{C} : \nabla_\Sigma \otimes_s \mathbf{u}_\Sigma, \quad \mathbf{T} = \kappa Gh(\nabla_\Sigma v - \boldsymbol{\theta}), \quad \mathbf{M} = \frac{h^3}{12} \mathbf{C} : \nabla_\Sigma \otimes_s \boldsymbol{\theta} \quad (\text{Eq. 10})$$

as functions of in-plane displacements $\mathbf{u}_\Sigma$, deflection v , and bending rotations $\boldsymbol{\theta}$. The constitutive equation for an isotropic elastic material writes:

$$\mathbf{C} : \boldsymbol{\varepsilon} = \frac{\nu E}{1-\nu^2} (\text{Tr} \boldsymbol{\varepsilon}) \mathbf{I}_d + \frac{E}{1+\nu} \boldsymbol{\varepsilon}$$

where $\mathbf{I}_d$ is the $d \times d$ identity matrix (here $d = 2$) and ∇_Σ is the gradient vector over the plate mid-surface. Equations of motion are:

$$\rho \partial_t^2 \mathbf{u}_\Sigma = \text{Div}_\Sigma \mathbf{N}, \quad \rho \partial_t^2 v = \text{div}_\Sigma \mathbf{T}, \quad \frac{\rho h^2}{12} \partial_t^2 \boldsymbol{\theta} = \text{Div}_\Sigma \mathbf{M} + \mathbf{T}. \quad (\text{Eq. 11})$$

Introducing $\mathbf{u} = (\partial_t \mathbf{u}_\Sigma, \partial_t v, \partial_t \boldsymbol{\theta}, \mathbf{N}, \mathbf{T}, \mathbf{M})$, (Eq. 10) and (Eq. 11) have the form of a system (Eq. 1) with symbol $\mathbf{P}(\mathbf{x}, \mathbf{k}) = i\mathbf{L}(\mathbf{x}, \mathbf{k})$ and $d = 2$, $n = 13$, where:

$$-\mathbf{L} = \begin{bmatrix} 0 & 0 & 0 & \rho^{-1} \mathbf{K}(\mathbf{k}) & \rho^{-1} \mathbf{m}(\mathbf{k}) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \rho^{-1} \mathbf{k}^\top & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & h(\rho I)^{-1} \mathbf{K}(\mathbf{k}) & h(\rho I)^{-1} \mathbf{m}(\mathbf{k}) \\ \mathbf{C} \mathbf{u} \mathbf{K}(\mathbf{k}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ G h \mathbf{m}(\mathbf{k})^\top & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \kappa G h \mathbf{k} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & D \mathbf{u} \mathbf{K}(\mathbf{k}) & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & G / \mathbf{m}(\mathbf{k})^\top & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (\text{Eq. 12})$$

$I = h^3/12$, $C = Eh/(1-\nu^2)$, $D = EI/(1-\nu^2)$, $\mathbf{k} = (k_1, k_2)$ and:

$$\mathbf{u} = \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix}, \quad \mathbf{K}(\mathbf{k}) = \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix}, \quad \mathbf{m}(\mathbf{k}) = (k_2, k_1).$$

Let $c_S^2 = Gh/\rho$, $c_T^2 = \kappa c_S^2$ and $c_P^2 = C/\rho$. The eigenvalues of the dispersion matrix $\mathbf{L}$ are now:

$$\begin{aligned} \lambda_T^\pm &= \pm c_T |\mathbf{k}| \quad \text{each with multiplicity 1,} \\ \lambda_S^\pm &= \pm c_S |\mathbf{k}| \quad \text{each with multiplicity 2,} \\ \lambda_P^\pm &= \pm c_P |\mathbf{k}| \quad \text{each with multiplicity 2,} \end{aligned} \quad (\text{Eq. 13})$$

and $\lambda_0 = 0$ with multiplicity 3. Modes 0 are non propagative vortical modes which may be disregarded, mode T corresponds to a pure transverse shear energy, and the double modes P and S correspond to in-plane and bending energies. These results have been extended to thick curved shells in [4].

POWER FLOW REFLECTION/TRANSMISSION COEFFICIENTS IN ASSEMBLIES

Boundary conditions for the power flows as imposed by the boundary conditions for displacements and/or boundary forces have been ignored so far. Their rigorous characterization is an open mathematical problem to date. However, a formal derivation of high-frequency power flow reflection/transmission coefficients in coupled beams (figure 1) or plates (figure 2) can be proposed based on the previous developments for their energy evolution properties.

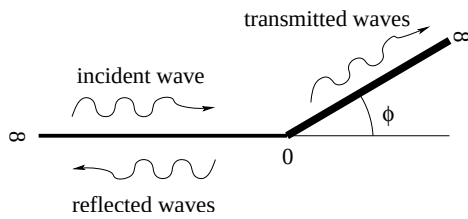


Figure 1.-Two coupled beams

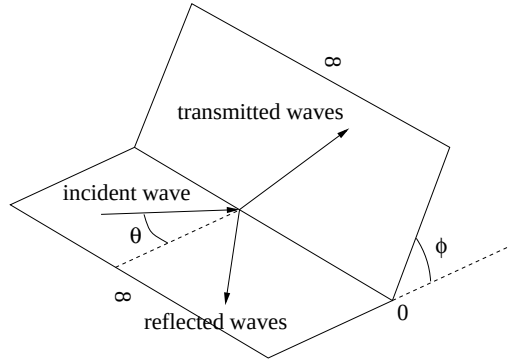


Figure 2.-Two coupled plates

Reflection/transmission coefficients for coupled beams

Let us consider two semi-infinite beams defined by their curvilinear abscissa $s \in \mathbb{R}_-$ for beam #1 and $s \in \mathbb{R}_+$ for beam #2. ϕ is the angle of the junction (see figure 1). For $j = 1$ or 2, let ρ_j be the mass per unit length of beam j , I_j its cross-section inertia, S_j its cross-section area, ν_j its Poisson ratio, E_j its Young's modulus, and $G_j' = \kappa_j G_j$ its reduced shear modulus where κ_j is the usual shear reduction coefficient which depends on the beam cross-section geometry and Poisson ratio. For plane waves propagating in the beams at the circular frequency ω , one introduces the wave numbers $k_{\alpha j}(\omega) = \omega/c_{\alpha j}$, with $\alpha \in \{P, T\}$.

Given the above results for the spectrum of the dispersion matrix (Eq. 8) of a beam, we assume that « high-frequency » waves propagating in coupled thick beams have the following form. For a pure shear motion ($\alpha = T$), the incident wave is $v_i(s) = e^{-ik_{T1}s}$, or $\theta_i(s) = -ik_{P1}e^{-ik_{P1}s}$ for a pure bending motion ($\alpha = Pb$), or $u_i(s) = e^{-ik_{P1}s}$ for a longitudinal motion ($\alpha = Pn$). The reflected waves in beam #1 are written:

$$v_r(s) = Ae^{ik_{T1}s}, \quad \theta_r(s) = ik_{P1}Be^{ik_{P1}s}, \quad u_r(s) = Fe^{ik_{P1}s}, \quad (\text{Eq. 14})$$

and the transmitted waves in beam #2 are written:

$$v_t(s) = He^{-ik_{T2}s}, \quad \theta_t(s) = -ik_{P2}Je^{-ik_{P2}s}, \quad u_t(s) = Le^{-ik_{P2}s}. \quad (\text{Eq. 15})$$

Coefficients A, B, F, H, J et L are obtained from the interface conditions at the junction $s = 0$, that is $(u, v, \theta)|_{s=0}$ and $(N, T, M)|_{s=0}$ are continuous across the junction. The corresponding power flows are derived from the generic formula of the energy flux associated to plane waves propagating in a beam:

$$\Pi = \frac{1}{2} \text{Re} \{ i\omega (N\bar{u} + T\bar{v} + M\bar{\theta}) \}. \quad (\text{Eq. 16})$$

The power reflection coefficients $\rho_{\alpha\beta}^{11}$ for beam #1, and the power transmission coefficients $\tau_{\alpha\beta}^{12}$ for beam #2 are computed by:

$$\rho_{\alpha\beta}^{11}(\phi) = -\frac{\Pi_{\alpha t}}{\Pi_{\beta i}}, \quad \tau_{\alpha\beta}^{12}(\phi) = \frac{\Pi_{\alpha t}}{\Pi_{\beta i}}, \quad \alpha, \beta \in \{Pn, Pb, T\}, \quad (\text{Eq. 17})$$

where Π_i, Π_r and Π_t stand for the incident, reflected and transmitted fluxes, respectively (– signs correspond to a reversed propagation direction).

Reflection/transmission coefficients for coupled plates

Let us consider two semi-infinite thick plates defined by their local coordinate systems (x, y) such that $x \in \mathbb{R}_-$ for plate #1 and $x \in \mathbb{R}_+$ for plate #2. Their junction is the line $\{x = 0\}$ and ϕ is its angle (see figure 2). For $j = 1$ or 2 , let ρ_j be the mass per unit surface of plate j , h_j its thickness, ν_j its Poisson ratio, E_j its Young's modulus, and $G_j' = \kappa_j G_j$ its reduced shear modulus. For plane waves propagating in the plates at the circular frequency ω , one introduces the wave numbers $k_{\alpha j}(\omega) = \omega/c_{\alpha j}$, with $\alpha \in \{P, S, T\}$, such that $k_{\alpha j}^2 = k^2 + \chi_{\alpha j}^2$, where $k \in \mathbb{R}$ is the tangent wave number of the incident wave on the junction. It remains unchanged by the reflection/transmission process on the junction line by Snell-Descartes law. If θ is the incidence angle of a wave α in plate #1 with respect to the normal to the junction line (see figure 2), then $k = k_{\alpha 1} \sin \theta$. In return, the normal wave numbers $\chi_{\alpha j}$ are either real (propagative waves), or purely imaginary (evanescent waves localized on the junction line) depending on the value of k with respect to $k_{\alpha j}$.

Given the above results for the spectrum of the dispersion matrix (Eq. 12) of a plate, we assume that « high-frequency » waves propagating in coupled thick plates have the following form. For pure shear motion ($\alpha = T$), the incident wave is:

$$v_i(x, y) = e^{-ik_{T1}(x \cos \theta + y \sin \theta)}, \quad k = k_{T1} \sin \theta, \quad (\text{Eq. 18})$$

for pure bending motion ($\alpha = Pb, Sb$) it is:

$$\theta_{Pi}(x, y) = -ik_{P1} \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} e^{-ik_{P1}(x \cos \theta + y \sin \theta)}, \quad \theta_{Si}(x, y) = -ik_{S1} \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} e^{-ik_{S1}(x \cos \theta + y \sin \theta)}, \quad (\text{Eq. 19})$$

and for in-plane motions ($\alpha = Pn, Sn$) it is:

$$\mathbf{u}_{Pi}(x, y) = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} e^{-ik_{P1}(x \cos \theta + y \sin \theta)}, \quad \mathbf{u}_{Si}(x, y) = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} e^{-ik_{S1}(x \cos \theta + y \sin \theta)}, \quad (\text{Eq. 20})$$

with $k = k_{\alpha 1} \sin \theta$ in both cases. The reflected waves in plate #1 are written:

$$v_r(x, y) = Ae^{i(\chi_{T1}x - ky)} \quad (\text{Eq. 21})$$

for pure shear motion,

$$\theta_r(x, y) = iB \begin{pmatrix} -\chi_{P1} \\ k \end{pmatrix} e^{i(\chi_{P1}x - ky)} + iF \begin{pmatrix} k \\ \chi_{S1} \end{pmatrix} e^{i(\chi_{S1}x - ky)} \quad (\text{Eq. 22})$$

for pure bending motion, and:

$$\mathbf{u}_r(x, y) = \frac{H}{k_{P1}} \begin{pmatrix} -\chi_{P1} \\ k \end{pmatrix} e^{i(\chi_{P1}x - ky)} + \frac{J}{k_{S1}} \begin{pmatrix} k \\ \chi_{S1} \end{pmatrix} e^{i(\chi_{S1}x - ky)} \quad (\text{Eq. 23})$$

for in-plane motions. The transmitted waves in plate #2 are written:

$$v_t(x, y) = Ke^{-i(\chi_{T2}x + ky)} \quad (\text{Eq. 24})$$

for pure shear motion,

$$\theta_t(x, y) = -iL \begin{pmatrix} \chi_{P2} \\ k \end{pmatrix} e^{-i(\chi_{P2}x + ky)} - iM \begin{pmatrix} -k \\ \chi_{S2} \end{pmatrix} e^{-i(\chi_{S2}x + ky)} \quad (\text{Eq. 25})$$

for pure bending motion, and:

$$\mathbf{u}_t(x, y) = \frac{N}{k_{P2}} \begin{pmatrix} \chi_{P2} \\ k \end{pmatrix} e^{-i(\chi_{P2}x + ky)} + \frac{O}{k_{S2}} \begin{pmatrix} -k \\ \chi_{S2} \end{pmatrix} e^{-i(\chi_{S2}x + ky)} \quad (\text{Eq. 26})$$

for in-plane motions. Coefficients $A, B, F, H, J, K, L, M, N$ et O are obtained from the interface conditions at $x = 0$, that is $(\mathbf{u}, \mathbf{v}, \boldsymbol{\theta})|_{x=0}$ and $(\mathbf{N}\hat{\mathbf{n}}, \mathbf{T} \cdot \hat{\mathbf{n}}, \mathbf{M}\hat{\mathbf{n}})|_{x=0}$ are continuous across the junction

line of which unit normal is $\hat{\mathbf{n}}$. The corresponding power flows are derived from the generic formula of the energy flux associated to plane waves propagating in a plate:

$$\mathbf{\Pi} = \frac{1}{2} \text{Re} \{ i\omega (\mathbf{N}\bar{\mathbf{u}} + \mathbf{T}\bar{\mathbf{v}} + \mathbf{M}\bar{\boldsymbol{\theta}}) \}. \quad (\text{Eq. 27})$$

The power reflection coefficients $\rho_{\alpha\beta}^{11}$ for plate #1, and the power transmission coefficients $\tau_{\alpha\beta}^{12}$ for plate #2 are computed by:

$$\rho_{\alpha\beta}^{11}(\theta, \phi) = -\frac{\mathbf{\Pi}_{\alpha r} \cdot \hat{\mathbf{n}}}{\mathbf{\Pi}_{\beta i} \cdot \hat{\mathbf{n}}}, \quad \tau_{\alpha\beta}^{12}(\theta, \phi) = \frac{\mathbf{\Pi}_{\alpha t} \cdot \hat{\mathbf{n}}}{\mathbf{\Pi}_{\beta i} \cdot \hat{\mathbf{n}}}, \quad \alpha, \beta \in \{Pb, Pn, Sb, Sn, T\}, \quad (\text{Eq. 28})$$

where $\mathbf{\Pi}_i, \mathbf{\Pi}_r$ and $\mathbf{\Pi}_t$ stand for the incident, reflected and transmitted fluxes, respectively (– signs correspond to a reversed propagation direction with respect to $\hat{\mathbf{n}}$).

The high-frequency power flow reflection coefficients of a semi-infinite single beam and a semi-infinite single plate for either Dirichlet ($E_1/E_2 \rightarrow 0$) or Neumann ($E_2/E_1 \rightarrow 0$) boundary conditions at their end are directly obtained from these results. Computations show that they are all equal to 1 without mode conversion of any type for a beam. In-plane Pn/Sn and Sn/Pn mode conversions occur for power flows in a plate, similarly to polarized elastic waves, with the emergence of a critical incidence angle $\theta_c = \text{Arc sin} \sqrt{(1-\nu)/2}$ for Sn/Pn reflection.

NUMERICAL EXAMPLE

An assembly of two thick plates with $\phi = 60^\circ$ is considered in this example. Plate #1 is impacted at $t = 0$ by a mechanical shock close to the junction line which initially loads shear mode T . Liouville transport equations are solved by a direct Monte-Carlo method [2], and the energy rays for modes $\alpha = T, S$ or P are reflected/transmitted by the junction according to the scattering laws derived previously. Both plates have identical parameters with $\nu_1 = \nu_2 = 0.2$. The evolution of the energy density of (Eq. 4) is displayed on figure 3, where $\tau = L/c_T$ and L is the width of the plates.

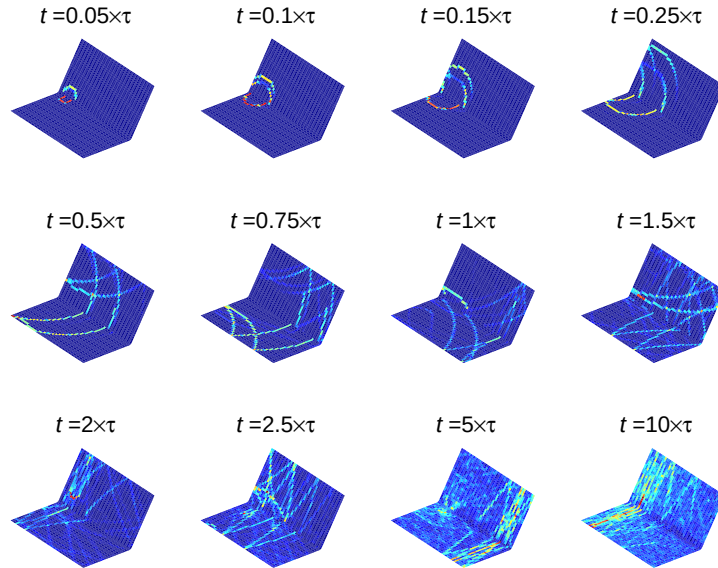


Figure 3.-Evolution of mechanical energy density in a plate assembly impacted by a shear force

References:

- [1] P. Gérard, P. A. Markowich, N. J. Mauser, F. Poupaud: Homogenization limits and Wigner transforms. Communications on Pure and Applied Mathematics **L** (1997) 323–379
- [2] B. Lapeyre, E. Pardoux, R. Sentis: Méthodes de Monte-Carlo pour les Équations de Transport et de Diffusion. Springer-Verlag, Berlin (1998)
- [3] P.-L. Lions, T. Paul: Sur les mesures de Wigner. Revista Matemática Iberoamericana **9** (1993) 553-618
- [4] É. Savin: Transient transport equations for high-frequency power flow in heterogeneous cylindrical shells. Waves in Random Media **14** (2004) 303-325